

Ans. Q.55

A leap year has 366 days = 52 weeks and 2 extra days.

These 2 consecutive days can be: (Sun, Mon), (Mon, Tue), (Tue, Wed), (Wed, Thu), (Thu, Fri), (Fri, Sat), (Sat, Sun).

Total possible outcomes = 7. Favourable outcomes containing Monday = {(Sun, Mon), (Mon, Tue)} = 2.

P(53 Mondays) = $\frac{2}{7} \rightarrow$ Correct Option: (a)

Ans. Q.56

Two digit numbers start from 10 to 99. Total two-digit numbers = $99 - 10 + 1 = 90$.

Multiples of 3 are: 12, 15, 18, ..., 99. Number of multiples = 30.

Probability = $\frac{30}{90} = \frac{1}{3}$.

P(Multiple of 3) = $\frac{1}{3} \rightarrow$ Correct Option: (c)

Ans. Q.1

Given: 5 black, 7 red, 3 white balls. Total balls $n(S) = 5 + 7 + 3 = 15$ balls.

(i) Red:

Favourable = 7

$P(\text{Red}) = \frac{7}{15}$

(ii) Black or White:

Favourable = $5 + 3 = 8$

$P(\text{Black or White}) = \frac{8}{15}$

(iii) Not Black:

Favourable = $7 + 3 = 10$

$P(\text{Not Black}) = \frac{10}{15} = \frac{2}{3}$

Ans. Q.2

Given: 7 black, 5 red, 3 white balls. Total balls $n(S) = 7 + 5 + 3 = 15$ balls.

(i) Red:

Favourable = 5

$P(\text{Red}) = \frac{5}{15} = \frac{1}{3}$

(ii) Black or White:

Favourable = $7 + 3 = 10$

$P(\text{B or W}) = \frac{10}{15} = \frac{2}{3}$

(iii) Not Black:

Favourable = $5 + 3 = 8$

$P(\text{Not Black}) = \frac{8}{15}$

Ans. Q.3

Total number of balls = 6 red + 5 black + 4 white = 15 balls. Total outcomes = 15.

(i) P(white) = $\frac{4}{15}$

(ii) P(red) = $\frac{6}{15} = \frac{2}{5}$

(iii) P(not black) = $(6 + 4) / 15 = \frac{10}{15} = \frac{2}{3}$

(iv) P(red or white) = $(6 + 4) / 15 = \frac{10}{15} = \frac{2}{3}$

Ans. Q.4

Total cards in a standard deck = 52.

Number of aces = 4, Number of kings = 4.

Cards that are either an ace or a king = $4 + 4 = 8$.

Number of cards which are neither ace nor king = $52 - 8 = 44$.

P(neither ace nor king) = $\frac{44}{52} = \frac{11}{13}$

Ans. Q.5

Total cards = 52. Red cards = 26.

Black kings = 2.

Red kings already counted in red cards

Cards to be excluded = 26 red cards + 4 kings = 30 cards.

Remaining cards = $52 - 30 = 22$ cards.

P(neither red card nor black king) = $\frac{22}{52} = \frac{11}{26}$

Ans. Q.6

Total number of bulbs = 400. Defective bulbs = 15.

Number of non-defective bulbs = $400 - 15 = 385$.

$P(\text{bulb is not defective}) = \frac{385}{400}$.

Dividing numerator and denominator by 5: $P = \frac{77}{80}$.

Ans. Q.7

A leap year contains 366 days = 52 complete weeks + 2 extra odd days.

52 weeks will definitely contain 52 Fridays

Possible pairs for the 2 extra days: { (Sun, Mon), (Mon, Tue), (Tue, Wed), (Wed, Thu), (Thu, Fri), (Fri, Sat), (Sat, Sun) }

→ Total outcomes = 7.

Favourable outcomes with Friday = { (Thu, Fri), (Fri, Sat) } = 2.

$P(\text{53 Fridays}) = \frac{2}{7}$.

Ans. Q.8

Total pack of cards = 52. Red cards = 26. Queens = 4. **Red queens already counted in red cards**

Unfavourable cards = $26 - 4 = 30$ cards.

Favourable cards = $52 - 30 = 22$ cards.

$P(\text{neither a red card nor queen}) = \frac{22}{52} = \frac{11}{26}$.

Ans. Q.9

Numbers from 1 to 25. Total outcomes $n(S) = 25$. **1 is not a prime number**

Prime numbers between 1 and 25 are: 2, 3, 5, 7, 11, 13, 17, 19, 23 → Total primes = 10.

Therefore, numbers which are not prime = $25 - 10 = 15$.

$P(\text{not a prime number}) = \frac{15}{25} = \frac{3}{5}$.

Ans. Q.10

Cards are numbered 1 to 30. Total outcomes = 30.

Numbers divisible by 3 are: {3, 6, 9, 12, 15, 18, 21, 24, 27, 30}. Total count = 10.

Numbers not divisible by 3 = $30 - 10 = 20$.

$P(\text{not divisible by 3}) = \frac{20}{30} = \frac{2}{3}$.

Ans. Q.11

Given: 5 red, 8 white, 7 black balls. Total balls $n(S) = 5 + 8 + 7 = 20$ balls.

(i) Red or White:

Favourable = $5 + 8 = 13$

$P = \frac{13}{20}$

(ii) Not Black:

Favourable = $20 - 7 = 13$

$P = \frac{13}{20}$

(iii) Neither White nor Black:

Only red balls = 5

$P = \frac{5}{20} = \frac{1}{4}$

Ans. Q.12

Cards numbered 1 to 20. Total outcomes = 20.

(i) Even:

{2, 4, 6, 8, 10, 12, 14, 16, 18, 20} = 10

$P(\text{Even}) = \frac{10}{20} = \frac{1}{2}$

(ii) Prime:

{2, 3, 5, 7, 11, 13, 17, 19} = 8

$P(\text{Prime}) = \frac{8}{20} = \frac{2}{5}$

(iii) Multiple of 3:

{3, 6, 9, 12, 15, 18} = 6

$P(\text{Mult 3}) = \frac{6}{20} = \frac{3}{10}$

Ans. Q.13

Cards numbered 1 to 25. Total outcomes = 25.

(i) Even:

(ii) Prime:

(iii) Multiple of 6: